

MIL-STD-29A

MILITARY STANDARD

DRAWING REQUIREMENTS FOR MECHANICAL SPRINGS

1. SCOPE

1.1 SCOPE. This standard illustrates the approved method of drawing, dimensioning, and specifying spring data on detail drawings of each type of mechanical spring described herein. Adherence to this standard is mandatory in the preparation of drawings prepared by or for the Department of Defense. Appendix A contains material, design and manufacturing data representative of modern spring engineering. It is included herein for guidance and convenience only and contains no mandatory provisions.

1.1.1 Types of Springs. The types of springs described in this standard are:

1.1.1.1 Compression.

(a) Helical, Cylindrical; (b) Helical, Stranded Wire; (c) Volute; (d) Coned Disc (Belleville)

1.1.1.2 Extension.

(a) Helical

1.1.1.3 Torsion.

(a) Helical; (b) Torsion bar; (c) Spiral

1.1.1.4 Flat.

(a) Cantilever

1.1.1.5 Constant force.

(a) Flat Strip

1.1.1.6 Garter.

(a) Helical

1.2 PURPOSE. The purpose of this standard is to establish uniform methods of specifying end product data for mechanical

springs on drawings prepared by or for the Department of Defense.

2. REFERENCED DOCUMENTS

2.1 NOT APPLICABLE

3. DEFINITIONS

3.1 MECHANICAL SPRING. An elastic body whose mechanical function is to store energy when deflected under load and return the equivalent amount of energy upon being released.

3.2 LOAD. The force exerted upon or by a spring in order to reproduce or modify motion or to maintain a force system in equilibrium. A static load is slowly applied and does not subject a spring to fatigue. A dynamic load is either a suddenly applied load or one that causes impact due to kinetic energy. It also may be frequently repeated, thus shortening fatigue life.

3.3 TOTAL DEFLECTION. The movement of a spring from free position to its maximum operating position. In a compression spring, it is the deflection from the Free Length to the Solid Length.

3.4 DEFLECTION PER COIL. The total deflection of the spring divided by the number of active coils.

3.5 SET. The permanent distortion from the manufactured dimensions which occurs when the spring is stressed beyond the elastic limit of the material.

3.6 ACTIVE COILS. The number of coils which are used in computing the total deflection of the spring.

3.7 TOTAL COILS. The number of active

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coils plus the coils forming the ends, (Compression Springs).

3.8 SOLID LENGTH. The overall length of a compression spring when all coils are fully compressed.

3.9 FREE LENGTH. The overall length of a spring in the unloaded position.

3.10 PITCH. The distance between centers of adjacent active coils of a spring in the unloaded position.

3.11 INITIAL TENSION. A load wound into certain helical extension springs during the coiling operation which keeps the coils tightly closed and which must be exceeded by an applied load before the coils begin to open.

3.12 SPRING INDEX. The ratio of the mean spring diameter to the diameter of the wire.

$$\text{Spring index} = D/d = \frac{OD - d}{d} = \frac{ID + d}{d}$$

3.13 SPRING RATE. The load required to deflect a compression or extension spring one inch or the load required to deflect a torsion spring one degree or one revolution. Also referred to as "Scale," "Gradient" and "Load Factor."

3.14 TORQUE. A turning force about an axis multiplied by the distance from the load to the axis, usually expressed in pound-inches (LB IN.) or in ounce-inches (OZ IN.) and always used in conjunction with the number of degrees of rotation, number of revolutions or deflected position.

3.15 SPRING TOLERANCE. The permissible variations from a given dimension.

3.16 FACTOR OF SAFETY. The ratio of the maximum load a spring can sustain without permanent set to the maximum applied load.

3.17 DIRECTION OF HELIX. When a spring

is viewed from one end, the direction of helix is right hand when the coil recedes in a clockwise direction, and left hand when it recedes in a counterclockwise direction. A right hand helix follows the same direction as the threads on a standard screw. A left hand helix is most popular for compression and extension springs.

3.18 REFERENCE DIMENSION. A reference dimension is a dimension without tolerance which is entered on a drawing for informational purposes and which does not directly govern manufacturing or inspection operations. Reference dimensions are indicated on drawings by writing the abbreviation REF directly following or under the dimension.

3.19 STRESS RELIEVE. To subject springs to a low temperature heat treatment after coiling or bending to remove residual stresses caused by the forming operation. Also called "strain relieve", "stress equalizing", "tempering", "blueing" and "baking".

3.20 RANGE OF STRESS. The difference between the stress at maximum and minimum loads.

3.21 HARDEN. To cause a material to become harder by heat treatment. Spring steels are heated above the critical temperature and quenched in oil. Precipitation hardening alloys such as Beryllium-Copper, K-Monel and Inconel X are heated at elevated temperatures for extended periods of time.

3.22 ENDURANCE LIMIT. The maximum stress or range of stress at which a material will operate indefinitely without failure. This limit for springs is usually determined at 10,000,000 cycles of deflection.

3.23 ELASTIC LIMIT. The maximum stress to which a material may be subjected without causing a permanent set. Also the stress beyond which a material must be subjected to form a coil, radius or bend.

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3.24 MODULUS OF RIGIDITY. The measure of elastic ability. A mathematical constant expressing the stiffness, rigidity, elasticity and flexibility of a material. Also called "modulus of elasticity" and "Young modulus".

4. GENERAL REQUIREMENTS

4.1 Not Applicable

5. DETAIL REQUIREMENTS

5.1 GENERAL. Figures 2(a) through 2(d), 4(a) through 4(d), 7, 10, 12, 13, 14, 16, 17 and 18 are Drawing Requirement Charts illustrating approved methods of drawing and dimensioning the various types of springs listed in paragraph 1.1.1 and include compilations of spring data to be used as applicable when prescribing manufacturing information on detail drawings of mechanical springs. The inclusion herein of these charts should not be interpreted as restricting the scope of spring characteristics to be specified on engineering drawings delineating springs. The specification of correct and complete manufacturing data is the responsibility of the cognizant Design Activity. Figures 19 thru 30 are examples of typical detail drawings of springs illustrating the application of the information included on the charts.

5.1.1 Delineation. The simplified drafting methods shown on the Drawing Requirement Charts and the typical detail drawings, e.g., Figures 2(a) and 19, are generally satisfactory for depicting helical springs.

5.2 SPECIFICATIONS APPLICABLE TO ALL SPRINGS:

5.2.1 Material specifications. Material Specifications are designated in the space provided in the title block on the drawing. When such space is inadequate enter therein "SEE NOTE" and describe the material requirements in a general note.

5.2.2 Diameter of wire. Wire diameter, or

width and thickness of material, shall be designated in decimals, or common fractions of an inch, as applicable, and unless otherwise specified, all dimensions apply before plating or applying finishes. Such dimensions are used to avoid confusion resulting from various gage numbering systems in current use. However, they should conform with the appropriate gage customarily used with the material specified, whenever possible. For oil tempered and corrosion resisting steels such diameters should conform with the U. S. Steel Wire Gage, which is the same as the Washburn & Moen Gage. For non-ferrous materials the American Wire Gage, which is the same as the Brown & Sharpe Gage, is used. Music Wire is drawn to the American Steel Wire and Music Wire Gage but other gages and intermediate sizes are available. Do not specify the name of the Gage used. Tolerances on diameter and thickness shall not be specified when they are covered in the appropriate material specifications or by general tolerance notes. Specify a tolerance on the width, as applicable. See Table I, Preferred Sizes of Spring Materials, in Appendix A, Section I.

5.2.3 Inspection Notes. Detail drawings of springs that are critical, including those springs subjected to critical conditions of temperature, stress or corrosive environments, shall include appropriate instruction notes applicable to controlling the manufacture within the desired environments. It is appropriate to specify those spring factors requiring inspection, the tests required, and whether all springs produced from the particular drawing shall be inspected. Consistent with Quality Control Procedure the size of inspection samples shall not be specified.

5.3 FORMAT FOR SPECIAL DATA NOTES. When notes prescribing special data are required on detail drawings of springs, it is recommended that the following format be used:

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- (a) Stress at°F ±°F for minutes, after coiling (or forming).
- (b) Hardness range Rc to
- (c) Squareness of ends in free position within°.
- (d) Cold set to solid.
- (e) Shot peen (give specification) to intensity.
- (f) Protective coating (give specification).
- (g) Non destructive inspection (give specification).
- (h) The body shall not camber more than IN. in its entire length.
- (i) Test over arbor dia IN.
- (j) To withstand temperatures of °F for hours with loss of load not to exceed%.
- (k) To withstand deflections from initial to final position with loss of load not to exceed.....%.

5.4 HELICAL COMPRESSION SPRINGS

5.4.1 Definition. A compression spring is an open-coil helical spring that offers resistance to a compressive force applied axially. Compression springs are generally made cylindrical in form, although other forms are used, such as conical, tapered, concave or convex, consistent with design requirements. Because of ready supply, wire of circular cross section is used whenever possible; however, where advantageous to design, wire of square or rectangular cross section may be utilized. Figure 1 illustrates several forms of helical compression springs.

5.4.2 Drawing Requirements for Helical Com-

pression Springs. Guide lines for specifying pertinent dimensional and load data on engineering drawings of helical compression springs, which satisfy specific design requirements and, simultaneously, allow maximum latitude in manufacturing, are provided herein, categorized as follows:

(1) *No Load Specified.* The design activity assumes responsibility for the load capacity of the spring. The manufacturer is required to furnish a spring meeting the dimensional data specified. The free length, the coil diameter, and the total number of coils are specified, each with a tolerance. The tolerance on the wire diameter conforms to the governing material specification. Figure 2(a) shows suggested data applicable to springs in the "no load" category.

(2) *One Load Specified.* This category has application when the spring is required to develop a load, within a specified tolerance, preferably at the initial assembled length. The spring, normally, is not subjected to further deflection in operation. The manufacturer is required to meet the load requirement, but the free length and the total number of coils are not restricted, being designated REF. The tolerance on the wire diameter conforms to the governing material specification and the manner of coil diameter callout is dependent upon spring function. Figure 2(b) shows suggested data applicable to springs in the "one load" category.

(3) *Two Loads Specified.* This category has application when the spring is required to develop a load, within a specified tolerance, at each of two definite compressed lengths, normally at the initial and the final operating positions in the assembly. The manufacturer is required to meet the load requirements, other spring characteristics being designated as described in paragraph 5.4.2(2). Figure 2(c) shows suggested data applicable to springs in the "two loads" category.

(4) *Spring Rate Specified.* This category

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has application in assemblies in which the spring rate is the most significant characteristic, for example, in calibrated scales. The manufacturer is required to meet a prescribed spring rate, within a specified tolerance, but the free length and the total number of coils are permitted to vary, being designated REF. The tolerance on the wire diameter conforms to the governing material specification and the manner of coil diameter callout is dependent upon the spring function. Additionally, when closer control of the load to be developed at the initial assembled length is desired, one load, with a tolerance, to be developed at the assembled length should be specified. Figure 2(d) shows suggested data applicable to springs in the "spring rate" category.

5.4.2.1 Coil Diameter. Depending upon the application of the spring, specify one of the following:

- (a) TO WORK OVER.....
IN. (MAX) DIA ROD
- (b) TO WORK IN
IN. (MIN) DIA BORE
- (c) ID, with tolerance

(d) OD, with tolerance

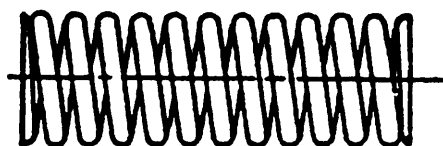
5.4.2.2 Direction of Helix. When governed by design requirements, specify the direction of helix as "LEFT HAND" (or LH), or "RIGHT HAND" (or RH), as applicable; otherwise specify as OPTIONAL. In most cases the hand is not important except when a plug screws into an end; or when one spring fits inside another, in which case one spring should be designated left hand and the other right hand.

5.4.2.3 Type of Ends. The types of ends having application to helical compression springs are illustrated in Figure 3. Specify the type of ends on the drawing; when necessary, the ends should be dimensioned.

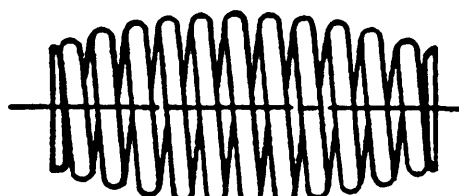
5.4.2.4 Solid Length. The maximum solid length shall be specified for any spring category when this parameter is essential to design requirements. However, solid length callout should be omitted whenever practicable. Except when necessary to satisfy function, springs should not be designed to go solid in operation.

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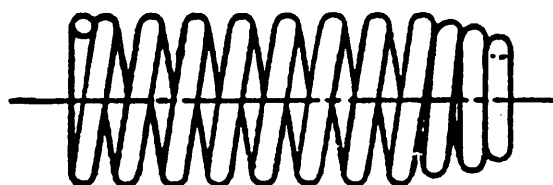
HELICAL COMPRESSION SPRING FORMS



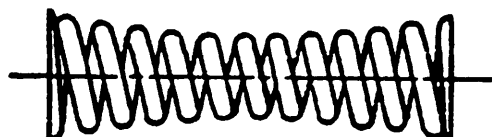
CYLINDRICAL
RIGHT HAND HELIX



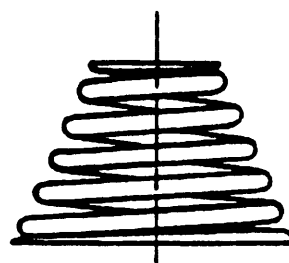
CONVEX
RIGHT HAND HELIX



CYLINDRICAL WITH CONED END
LEFT HAND HELIX



CONCAVE
RIGHT HAND HELIX

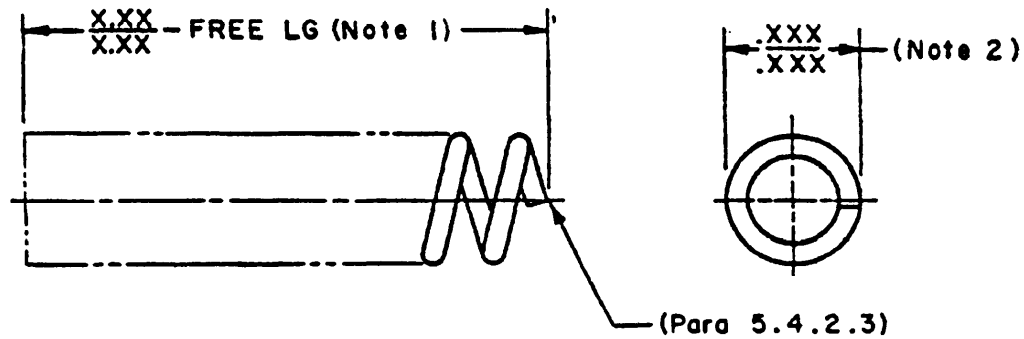


CONICAL
RIGHT HAND HELIX

FIGURE 1

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DRAWING REQUIREMENTS FOR HELICAL COMPRESSION SPRINGS
WHEN NO LOAD IS SPECIFIED



Spring data

	<i>Para</i>
MATERIAL SPECIFICATION	5.2.1
WIRE DIA	XXX IN. 5.2.2
DIRECTION OF HELIX	5.4.2.2
TOTAL COILS $XX \pm XX$	Note 3
SPECIAL DATA	5.3

Note 1: Specify free length, with tolerance (or by limits). (Refer to Appendix A, Section III, Table XIX.)

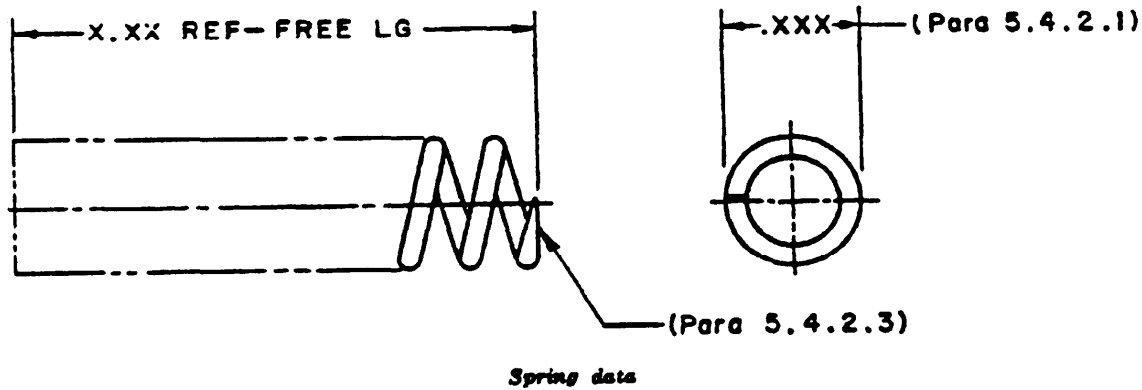
Note 2: When spring operates over guide, specify inside diameter of coil, with tolerance (or by limits). For other applications, specify outside diameter with tolerance (or by limits). (Refer to Appendix A, Section III, Table XVI.)

Note 3: Specify total number of coils, with tolerance. (Refer to Appendix A, Section III, Table XVII.)

FIGURE 2(a)

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DRAWING REQUIREMENTS FOR HELICAL COMPRESSION SPRINGS
WHEN ONE LOAD IS SPECIFIED



	Para
MATERIAL SPECIFICATION	5.2.1
WIRE DIA	XXX IN. 5.2.2
DIRECTION OF HELIX	5.4.2.2
TOTAL COILS	XX REF Note 1
LOAD AT COMPRESSED LG OF XXX IN. - XX LB ± XX LB	Note 2
SPECIAL DATA	5.3

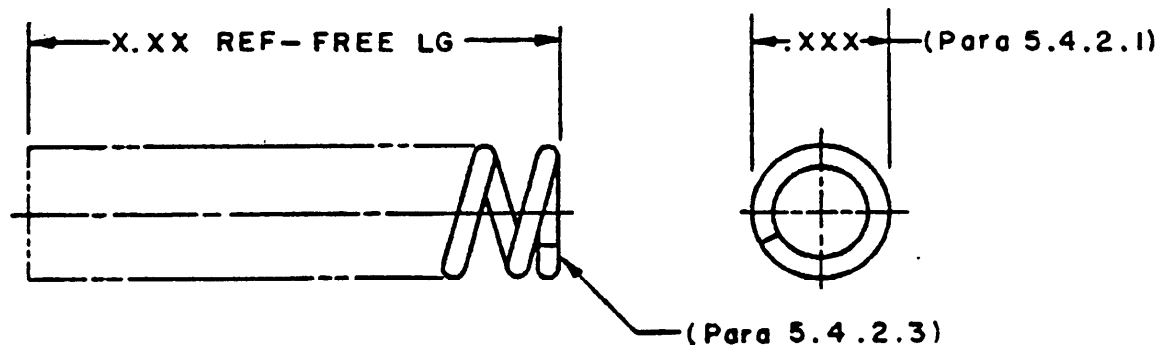
Note 1: Specify total number of coils; designate REF.

Note 2: Specify the one load, with tolerance, to be developed at a definite length, preferably the initial assembled length.

FIGURE 2(b)

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DRAWING REQUIREMENTS FOR HELICAL COMPRESSION SPRINGS
WHEN TWO LOADS ARE SPECIFIED



Spring data

	<i>Para</i>
MATERIAL SPECIFICATION	5.2.1
WIRE DIA	XXX IN. 5.2.2
DIRECTION OF HELIX	5.4.2.2
TOTAL COILS	XX REF Note 1
LOAD AT COMPRESSED LG OF XXX IN - XX LB $\pm$ XX LB	Note 2
LOAD AT COMPRESSED LG OF XXX IN - XX LB $\pm$ XX LB	Note 2
SPECIAL DATA	5.3

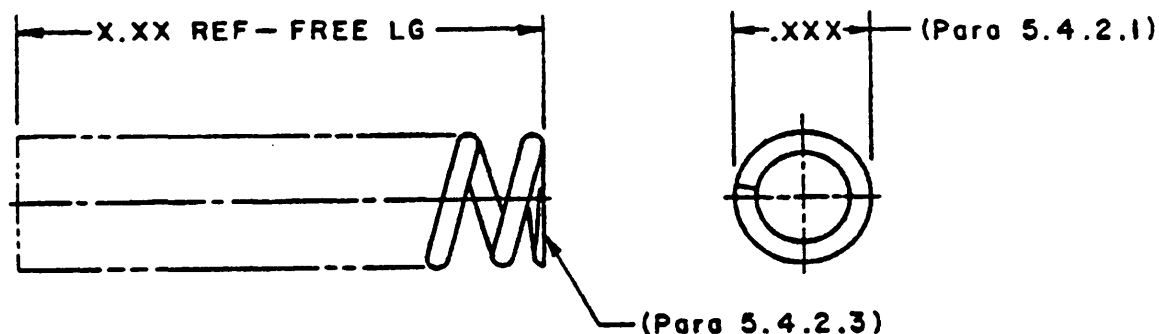
Note 1: Specify total number of coils; designate REF.

Note 2: Specify load to be developed at each of two definite compressed lengths, preferably at the initial and the final operating positions in the assembly.

FIGURE 2(c)

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DRAWING REQUIREMENTS FOR HELICAL COMPRESSION SPRINGS
WHEN SPRING RATE IS SPECIFIED



Spring data

	<i>Para</i>
MATERIAL SPECIFICATION	5.2.1
WIRE DIA	XXX IN. 5.2.2
DIRECTION OF HELIX	5.4.2.2
TOTAL COILS	XX REF Note 1
LOAD AT COMPRESSED LG OF XXX IN. - XX LB $\pm$ XX LB	5.4.2(4)
SPRING RATE XX LB/IN. $\pm$ XX LB/IN. BETWEEN COMPRESSED LG OF XX IN. AND XX IN.	Note 2
SOLID LG WITHOUT PERMANENT SET	XX IN. 5.4.2.4
SPECIAL DATA	5.3

Note 1: Specify total number of coils; designate REF.

Note 2: Specify spring rate, with tolerance.

FIGURE 2(d)

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TYPES OF HELICAL COMPRESSION SPRING ENDS

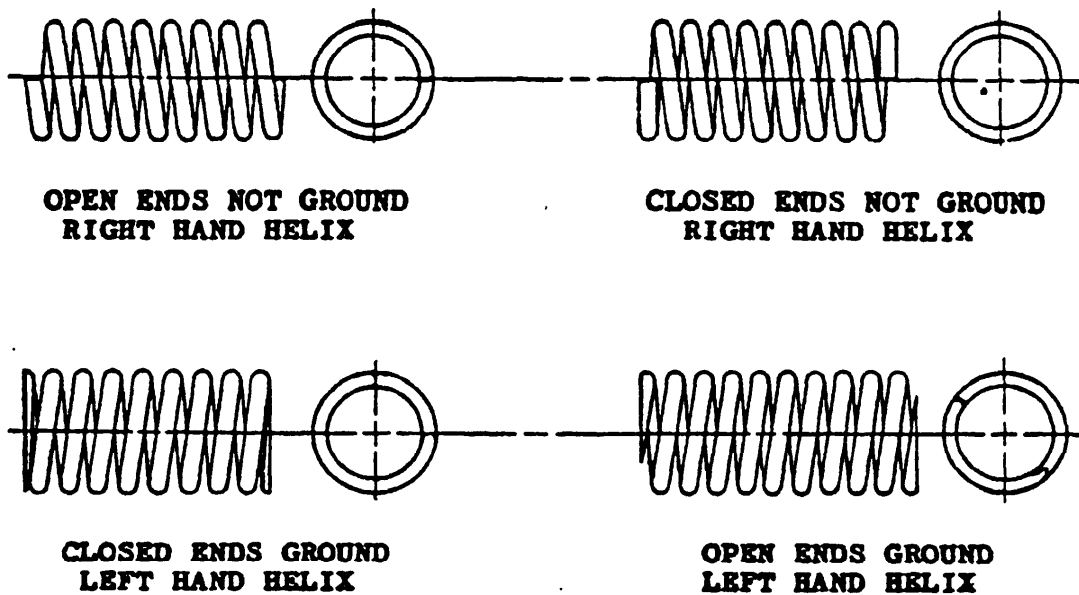


FIGURE 3

5.5 STRANDED WIRE COMPRESSION SPRINGS

5.5.1 Definition. Stranded wire compression springs are helical compression springs formed from two, three or more wires twisted about each other, or about one wire as a core, to form a single strand. Stranded wire compression springs have the advantage of damping the migratory waves that traverse the spring under shock loading. Such springs should be specified with caution since stranded wire is specially made and is not readily available.

5.5.2 Drawing Requirements for Stranded Wire Compression Springs. Figure 20 is a typical detail drawing of a stranded wire helical compression spring. The methods of drawing and dimensioning, and the spring data applicable to stranded wire springs correspond in general to those shown in Figures 2(c) for compression springs, except that the additional spring information noted herein is required on detail drawings:

- (a) Number of Wires in Strand.
- (b) Diameter of each Wire.
- (c) Length of Lay (Distance parallel to strand axis in which a single wire makes one turn.)
- (d) Direction of Strand Helix
NOTE: The strand helix shall be opposite in direction to the coil helix.
- (e) Direction of Coil Helix (Left Hand or Right Hand).
- (f) Type of Ends. Specify either CLOSED ENDS NOT GROUND or OPEN ENDS NOT GROUND, as applicable. Specify that ends are to be soldered, brazed or welded to prevent unraveling, and sharp burrs shall be removed.

5.6 HELICAL EXTENSION SPRINGS

5.6.1 Definition. A helical extension spring is a close-wound spring with or without initial tension, or an open-wound spring